Analog Code Notes

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This document provides notes on the analog program.

In the main() function, the program initializes analog to digital converter module 0, sets the sample input channel to AIN3 with hardware averaging of 4 samples. The main() function then enters an endless loop. In the loop, the program

1. Requests that the input be sampled and then reads the RAW 12-bit ADC output result
2. Converts the raw value to temperature, assuming that an LM60 temperature sensor is connected to AIN3
3. Filters the raw value from 1 with a sliding average FIR filter using a circular buffer
4. Converts the result from 3 to an FIR-filtered temperature
5. Filters the value from 2 with an IIR filter to give an IIR-filtered temperature

The initAdc0Ss3() function initializes ADC0 to sample at 1 Msps. Sequential sampler 3 is configured to use a SW trigger source and stop after one sample (this is all SS3 can do). Empirically, it was determined that a 16 clock delay after enabling clocks for ADC module 0 is needed to prevent the processor from faulting when the ADC0 registers are written.

The setAdc0Ss3Log2AverageCount() function determines the amount of hardware averaging that is performed. If hardware averaging is requested, then the sample time is dithered to improve the signal quality.

The setAdc0Ss3Mux() function determines which AIN pin is used as the signal input for SS3.

The readAdc0Ss3() function sends a SW trigger to SS3 and waits for the conversion to complete. If then returns the raw 12-bit ADC result,